Army - Signal Air Warning - AAA Air Warning - Battalions - Companies

Mobile Military Radar

Wednesday February 17. 2010

Notice:

Help !! Information, manuals, CDCs about IFF needed for web page.

Radar Descriptions

MPQ - Mobile; Radar; Special

MPQ-2 - Bomb scoring radar;
Notes: Used in Korea in 1951;

MPQ-4; A – Mortar Location
Frequency: 16,000 MHz Ku Band
Power Output: 80 kw peak
Pulse Width: 0.25 µseconds
Pulse Repetition Rate: 8600 pps
Range: 10 Km
Vertical Coverage:
Horizontal Coverage: 445 mil (25 degrees) fixed
Antenna: duel beam
Speed: rapid scan; non tracking
Beam Width:
Indicator Type and Quantity:
Manufacturer: General Electric
Notes: Entered service late 1958 - early 1959
Photos

-4A
The AN/MPQ-4A is a mobile, intercept-type (non-tracking) radar designed primarily to direct and locate hostile weapons and secondarily to adjust low-velocity artillery fire. Local or remote operation of the set is possible. In local operation, the control indicator group is operated in position on the radar trailer. In remote operation, the control indicator group is removed up to 45 meters from the radar trailer.
Range: 15,000 meters maximum, 225 meters minimum
Indicator: B-scope
Horizontal Coverage: 445 mil fixed-sector scanning
Frequency: 16 GHz ±160 MHz
Power Output: 50 kilowatts minimum
Pulse Repitition Rate: 7,000 pps
Pulse Width: 0.25 microseconds
Weight: 6,633 pounds, with 55 gallons of fuel
Manufacturer: General Electic

-4B
Manufacturer: General Electric

MPQ-10, -A – Mortar Location
Notes: The MPQ-10 is a lightweight, transportable radar set that is designed to locate and track mortar and artillery shells. The radar set can be used with associated recording equipment to locate the point of origin and point of impact of the missiles. The tracker mount contains radar set components mounted on a modified 40-millimeter gun carriage. A dipole antenna and parabolid reflector radiate rf pulses in a narrow beam. The radar set control actuates servo drives which position the antenna and rf beam in azimuth and elevation.
Frequency: 2740 to 2960 MHz
Power Output: 180 watts average, 200 kilowatts peak
Pulse Width: 0.8 µseconds
Pulse Repetition Rate: 1,000 pps
Range: 10,000 and 20,000 yards maximum and 500 yards minimum
Accuracy: 20 yards

Vertical Coverage: -125 mils to +1,540 mils
Horizontal Coverage: 6,400 mils, continuous; sector scan 200 to 800 mils, adjustable
Antenna: Dipole with 68 3/4 inch parabolid reflector
Speed: Reflector: 8 rpm maximum, dipole: 3,600 rpm
Beam Width: 5 degrees Horizontal, 5 degrees Vertical
Indicator Type and Quantity: Oscilloscope, B-scope (7-inch) 0 to 10,000 yard and 0 to 20,000 yard ranges; presents sector scanned. J-scope (3-inch); 2,000 yard range; dial indication of range, azimuth, and elevation.
Manufacturer: Sperry
Notes: 1951; Replaced by the MPQ-4 in 1958;
Notes: The technique of locating the point of origin or impact of a shell involves two steps. First, the general location of the trajectory is obtained by sector scanning through an azimuth angle of 800 mils or less while observing the B-scope. When the target echo is seen on the oscilloscope, the range mark appearing on the B-scope is moved quickly to the approximate target range by rotating the Slant Range hand wheel. Also, sector scanning is stopped and the radar mount is positioned by adjusting the Azimuth hand wheel to place the B-scope sweep trace over the target echo. Next the projectile is tracked automatically by observing the J-scope for the next round to be fired. When the projectile is fired, a pulse appears on the J-scope or near the range tracking gate. If the target is in the range tracking gate, throwing a switch causes the radar set to track the target automatically in range, azimuth, and elevation.
Photos

MPQ-12; C – Missile Tracker
Frequency: 2700 to 2900 MHz
Power Output: 250 kw
Pulse Width: 0.25 µseconds
Pulse Repetition Rate: 364-1707
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna: 10 foot parabolic dish
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Reeves
Notes: Modified version of the SCR-584

-12C
Frequency: 2700 to 2900 MHz
Pulse Repitition Rate: 1,000 pps
Manufacturer: Sperry

MPQ-14 – Course Directing
Manufacturer: Advance/Ultrasonic.
Notes: First used by USMC in Korea, September 1951;

MPQ-16 - Tracking
Manufacturer: Sperry

MPQ-18 – Missile Tracker
Frequency: 2650 to 2950 MHz
Power Output: 600 kw
Pulse Width:
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna: 10 foot parabolic dish

Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Reeves
Notes:

MPQ-19 - Tracking
Manufacturer: Sperry

MPQ-21 – Missile Tracker
Frequency: 6000 MHz
Power Output: 1 Mw
Pulse Width:
Pulse Repetition Rate: 1,000 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna: 10 foot parabolic dish
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Sperry
Notes:

MPQ-24 - Mortar Detector
Manufacturer: Sperry

MPQ-25 – Fire Control
Frequency: 2700 to 2900 MHz
Power Output: 210 kw
Pulse Width:
Pulse Repetition Rate: Variable
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: ITT Gilfillan
Notes: Trailer mounted; For Corporal Missile system.

MPQ-29 – Plotting and tracking radar
Manufacturer: Sperry
Photos

MPQ-31 – Tracking and Guidance;
Range: 4,000 Km
Manufacturer: Canoga

MPQ-32 – Detection and Tracking
Manufacturer: Sperry Rand/Sylvania
Antenna: Mechanical Scan
Notes: 1962.

MPQ-33 – Hawk Tracker/Illuminator;
Frequency: 10,000 to 10,250 MHz
Power Output: 125 watts
Pulse Width:
Pulse Repetition Rate:

Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Raytheon
Notes:

MPQ-34 – Hawk CW Acquisition
Frequency: 10,000 MHz
Power Output: 200 Watts
Pulse Width:
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Raytheon
Notes:

MPQ-35 – Acquisition Radar
Frequency: 1250 to 1350 MHz
Power Output: 450 kw
Pulse Width: 3 µseconds
Pulse Repetition Rate: 667 & 800 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Raytheon
Notes: Hawk Pulse Radar

MPQ-37 – Hawk Range Only Radar;
Manufacturer: Raytheon

MPQ-39 – Hawk Tracker Radar
Manufacturer: Raytheon.

MPQ-43 – Nike Hercules Acquisition Radar
Frequency: 1350 to 1450 MHz
Power Output:
Pulse Width: 6 µseconds
Pulse Repetition Rate: 417 to 438 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: General Electric

Notes: Fixed version known as HIPAR.

MPQ-44 –
Notes:

> *A mobile version of the HIPAR radar (AN/MPQ-44) was deployed in 1967. It is mounted on five trailers and includes all of the necessary power generating equipment to operate the entire Nike Hercules fire control system. The radar is designed to be used in the ATBM or EFS configuration and, like the fixed-site HIPAR, the mobile system also uses the presentation system of the Nike Hercules system. (Ref: US Army Air Defense Digest, 1972)*

MPQ-46 – Hawk Tracking Radar
Note: Photos

MPQ-48 – Hawk Continuous Wave Acquisition Radar.

MPQ-49 – Forward Area Alerting Radar (FAAR).
Range: 15 Km
Manufacturer: Sanders Associates/Sperry
Notes: Used with Vulcan and Chaparral systems as detection radar. Shelter mounted; know as Trackstar when mounted M-577A2 truck; also mounted on M561 Gamma Goat. The FAAR radar when mounted is known as the MPQ-49. When dismounted it is known as the TPQ-32. Used in support of the Chaparral/Vulcan Battalion.
Photos

MPQ-50 – Hawk Pulse Acquisition Radar;
Notes: Medium and high altitude with digital MTI;

MPQ-51 – Hawk Range Only Radar
Frequency: 15,500 to 17,500 MHz
Power Output: 120 kw
Pulse Width: 0.6 µseconds
Pulse Repetition Rate: 1600 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna: 4 foot parabolic dish antenna
  Speed:
  Beam Width:
Indicator Type and Quantity:
Manufacturer: Raytheon
Notes:

MPQ-53 – Patriot Radar
Description: The MPQ-53(V) phased-array radar and associated processor are the heart of the Patriot system. It is a multifunction, electronically scanned radar mounted on an M-860 trailer and towed by an M-Engagement Control Center.
Frequency: 4 - 6 GHz
Power Output: 100 kw
Pulse Width:
Pulse Repetition Rate:
Range: 37 nautical miles
Vertical Coverage:
Horizontal Coverage: 120 degrees (engagement sector 90 degrees)
Antenna:
  Speed:
  Beam Width:
Target capacity: 50 simultaneous targets

Missile control capacity: 9 in final engagement
Indicator Type and Quantity:
Dimensions:
Weight: 79,008 lbs
Length: 56.08 feet
Height 11.83 feet
Width: 9.42 feet
Notes: The MIM-104 Patriot program began in 1963 as the Army Air Defense System 1970 (AADS-70), an attempt to develop a supplement to the MIM-23B Improved HAWK and a replacement for the MIM-14B Nike-Hercules missile system. An initial development contract was awarded to Raytheon in May 1967, and the program was renamed Patriot in May 1976.
Manufacturer: Raytheon
Notes: Multi-function search and track aircraft and missile radar. Price $2.5 million. The Patriot system was designed to operate under all weather conditions and to be capable of destroying maneuvering aircraft at all altitudes. It can guide several missiles to attack multiple targets simultaneously in a severe electronic jamming environment. Microelectronic technology has been applied to the system for purposes of reliability. Component standardization helps ensure a good system readiness record.

The MPQ-53(V) features a lens array using an optical feed. Sum and difference patterns are separately optimized with a monopulse feed. The aperture is round and uses nearly 5,000 ferrite phase shifters. Four-bit flux-driven non-reciprocal ferrite phase shifters and waveguide-type radiators are located at both apertures. The system has separate arrays for missile guidance and IFF.

The radar face is an immediately recognizable characteristic of the system. A large phased-array face dominates the upper portion of the antenna unit face. It performs both surveillance and tracking for the radar. Below it is a nearly circular, 5,000-element phase shifter, and below that are two smaller phase-shifter arrays of 50 elements each. A row of 18 rectangular boxes divide the antenna face roughly in half, with access boxes and two slightly larger planar arrays (250 phase shifters) below it. At least one planar array is for the command-guidance and receive links of the missile.

Before an engagement, the radar is aligned to cover the most likely direction of attack. During an engagement, the beam is steered electronically in azimuth and elevation. The system has the ability to select and prioritize a single target from a mass formation based on a determination of the potential threat level.

The Patriot missile is fired from mobile launchers. Each Patriot battalion has six Patriot fire units with quad launchers, also known as batteries. Each battery has two fire platoons outfitted with a total of eight quad launchers (which can be expanded to 16 if necessary), an MPQ-53(V) radar, an MSQ-104(V) engagement control station, an electric power plant, and an antenna mast group.
History:

Month	Year	Major Development
May	1967	Contract awarded to Raytheon
May	1974	Engineering development begins
Oct	1978	Final DT/OT Phase II testing initiated
Jan	1980	DT/OT Phase II testing completed
May	1980	Production decision
Sep	1980	NATO announces Patriot to replace Nike/Hercules
Aug	1982	First Patriot unit activated
Oct	1983	The Netherlands becomes first export customer
Dec	1983	U.S.-West German Patriot/Roland deal made
Apr	1984	Patriot confirmed for ATM role
Late	1984	Patriot first deployed in Germany
	1985	IOC

	1989	PAC 1 first deployed to Europe
	1990-91	PAC 1 and PAC 2 deployed to Saudi Arabia
	1991	Patriot deployed to Operation Desert Storm
Aug	1996	Contract awarded for Patriot anti-cruise missile upgrade, Phase II
Feb	1998	Anti-cruise missile upgrade completed
3Q	FY98	PAC 3 contractor DT&E
1Q	FY00	Start of PDB-5 software improvements
3Q	FY00	PAC 3 FUE
4Q	FY01	PDB-5 software improvements completed, PAC 3 Missile FUE
Feb	2001	Joint Arrow-Patriot proof-of-concept demonstrations
Feb	2002	Patriot Radar Enhancement III completed
1Q	FY02	Initiation of PAC 3 Evolutionary Block Upgrades
1Q	FY03	PAC 3 Missile Block 02 production DAB
3Q	FY04	DAB IPR PAC 3/MEADS Combined Program
4Q	FY04	PAC 3 Missile Block 04 production
4Q	FY05	PAC 3 Missile IOC
4Q	FY07	PAC 3 Missile Block 06 production DAB

History (cont): The first operational Patriot battalion was deployed to Europe in November 1984 at Giessen, Germany. Another was deployed to Hanau. By 1992, the U.S. Army had 54 Patriot batteries deployed at nine sites in Europe. The German Patriots attained Initial Operational Capability in 1989.

Other countries using MPQ-53:

Egypt
Germany
Greece
The Netherlands
Japan
Kuwait
Israel
Saudi Arabia
Italy
South Korea
Switzerland
Taiwan.
Turkey
United Arab Emirates (UAE)
United Kingdom

Photos

MPQ-54 – Surveillance
Manufacturer: Sanders Associates/Sperry
Notes: FAAR radar used with Vulcan and Chaparral as detection radar

MPQ-55 – Hawk Continuous Wave Acquisition radar

MPQ-57 – Hawk High Power Illuminator

MPQ-62 – Hawk Continuous Wave Acquisition radar
Notes: For detecting low to medium altitude aircraft; Trailer Mounted;

MPQ-63 – Instrumentation
Frequency: 9300 to 10,000 MHz
Power Output: 30 kw
Pulse Width: 1 µsecond
Pulse Repetition Rate: 1 to 10,000 pps
Range:
Vertical Coverage:

Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: ITT Gilfillan
Notes:

MPQ-64 Sentinel - Forward Area Air Defense-Ground Based Sensor (FAAD-GBS)
Frequency: X-band
Range: 40 km
Vertical Coverage: 4 km
Horizontal Coverage: 360 degrees
Antenna: phased-array
Manufacturer: Raytheon; Prime Contractor: Hughes Aircraft (Fullerton, CA)
Notes: 3D Battlefield Radar; Mission is to alert Avenger, Bradley Linebacker or Stinger teams of aircraft, cruise missiles; brochure

MPQ-501 – Motor Locating.
Manufacturer: Raytheon
Notes: For Canada

MPQ-T3A – Multiple AAA/SAM Threat Simulator System,
Manufacturer: DRS Technologies

MPS - Mobile; Radar; Detecting, Range and Bearing, Search

MPS-1 – Early-warning Radar
Frequency: 1280 to 1350 MHz
Power Output:
Pulse Width:
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer:
Notes: Project 424C at Camp Evans Signal Laboratory; Canceled August 1943.

MPS-2
Notes: See SCR-582, Mark III. This unit was the SCR-582, Mark III with the Army/Navy Designator of MPS-2.

MPS-3 – Height Finder
Manufacturer: Raytheon

MPS-4 – Height Finder
Frequency: 6275 to 6575 MHz
Power Output: 140 kw peak
Pulse Width: 0.37 or 1.3µsec
Pulse Repetition Rate: Variable, 615 to 650 pps
Range: 80 miles
Vertical Coverage: 40,000 feet; -2 degrees to +20 degrees;
Horizontal Coverage:
Antenna:

Speed: Rotation 0 - 7 rpm; Vertical Scanning Rate: 1 - 1.5 scans per second
Beam Width:
Indicator Type and Quantity: one (1) five inch A scope, one (1) seven inch RHI scope
Manufacturer: Hazeltine
Notes:

MPS-7 – Surveillance; Long Range
Frequency: 1220 to 1350 MHz;
Power Output: 650 kw peak
Pulse Width: 3 or 6 µseconds
Pulse Repetition Rate: 200 or 400 pps (MTI available for 400 pps only)
Range: 325 Nautical Miles
Vertical Coverage: Up to 60,000 feet
Horizontal Coverage: 360 degrees
Antenna:
Speed: 5 rpm
Beam Width:
Indicator Type and Quantity: Seven PPI's
Manufacturer: Bendix
Notes: Mobile version of the FPS-3

MPS-8 – Height Finder
Frequency: 9230 to 9404 MHz
Power Output: 250 kw peak
Pulse Width: 0.5 or 2 µseconds
Pulse Repetition Rate: 530 pps
Range: 120 Nautical miles
Vertical Coverage: -5,000 to +60,000 feet (-2 to +23 degrees)
Horizontal Coverage:
Antenna:
Speed: Horizontal Automatic, 1/3 rpm; manual, 0 to 6 rpm; Vertical Scanning Rate: Automatic 60 cycles per minute, manual 30 cycles per minute
Beam Width:
Indicator Type and Quantity: One (1) RHI;
Manufacturer: RCA
Notes: FPS-4 is the fixed version and TPS-10D the transportable version of the MPS-8.

MPS-9 – Tracking
Frequency: 2700 to 2900 MHz
Power Output: 250 to 650 kw
Pulse Width: 0.8 µseconds
Pulse Repetition Rate: 300 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Reeves Instrument Corp.
Notes: **MPS-9** modified as **MSQ-2**, modified as **MSQ-39**, modified as **MSQ-46**;

MPS-11 – Surveillance Radar Set
The AN/MPS-11 Radar Set is a mobile, long-range search radar used as an early warning set. This set is also used as a permanently fixed radar identified as the AN/FPS-8. It is equipped with moving target indication (MTI) of the "Coherent video cancellation" type.
Capability: This radar set is capable of detecting aircraft in any azimuth at maximum ranges of 200 nautical miles and at maximum altitudes of /!0,000 feet. It can either be used to sector scan at any azimuth, or for continuous rotation for 360 degrees in both

forward or reverse directions. The AN/MPS-11 provides power and video connections for a video mapping unit (GPA-30);power, video, and antenna connections for identification equipment (e.g. GPX-18/UPX-6); power for telephone communications equipment; and power and trigger for a height finding radar set (e.g. AN/TPS-37 or AN/TPS-40).
Frequency: 1280 to 1350 MHz
Power Output: 1 MW peak, 1.08 kW average
Pulse Width: (2) µseconds
Pulse Repetition Rate: 360 pps (depending on quartz delay line used)
Range: 200 miles; MTI 150 miles;
Vertical Coverage: 30 degrees
Horizontal Coverage: 360 degrees, 40 to 120 degrees sector scan
Antenna:
Speed: Variable 0 to 10 rpm, CW or CCW; Position (handwheel controlled); Sector 0 to 120 degrees;
Beam Width:
Indicator Type and Quantity: PPI (Originally up to six (6) GPA-126 PPI's); ANG CRC's had 12 UPA-35 PPI's (1971 - 1980);
Weight: 79,790 pounds
Manufacturer: General Electric
Notes: Mobile version of the FPS-8; IFF: UPX-6 Transceiver, GPX-18B decoder (Transistorized version of the GPX-18A); GPA-30 Video Mapper; -11 version the antenna is mounted on a portable 30 foot tower; Trailer V-112
Photos

-11**A** Characteristics same as the MPS-11 with exceptions noted below
Notes: Antenna mounted on converted 40mm gun carriage; IFF: UPX-6 Transceiver, GPX-18B decoder (Transistorized version of the GPX-18A); GPA-30 Video Mapper; Trailer V-112; Photos; Technical Manual

MPS-14 – Height Finder
Frequency: 2700 to 2900 MHz
Power Output: 5 Mw
Pulse Width:
Pulse Repetition Rate:
Range: Azimuth 200 nautical miles, vertical -5,000 to +60,000 feet(-2 to +32 degrees)
Vertical Coverage: Vertical Scan rate: 20 to 30 cycles per minute
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity: Two (2) 12.5 inch RHI's
Manufacturer: General Electric
Notes: Mobile version of the FPS-6
Photos

MPS-16 Height Finder
Frequency: 5280 MHz
Power Output: 1 Mw
Pulse Width: 2.5 µseconds
Pulse Repetition Rate: 300 to 364 pps
Range: 200 Nautical miles
Vertical Coverage: 100,000 feet
Horizontal Coverage:
Antenna:
Speed: 1/8 rpm
Beam Width:
Indicator Type and Quantity:
Manufacturer: Avco
Notes: Photos

MPS-19 Tracking
Frequency: 2700 to 2900 MHz
Power Output: 500 kw
Pulse Width: 0.8 µseconds
Pulse Repetition Rate: 300 to 2,000 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Reeves
Notes:

MPS-20 (XW-1) - Height Finder
Used with: 412L system (Phase I configuration) April 1961
Manufacturer: Westinghouse Electric Corp

MPS-21 Surveillance
Frequency: 1250 to 1350 MHz
Power Output: 5 Mw
Pulse Width: 5 µseconds
Pulse Repetition Rate: 360
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: Sperry
Notes: "V" beam 3D radar

MPS-22 - Tracking

MPS-23 Surveillance, 3D radar
Manufacturer: Hughes Aircraft Corp.
Notes:

> *A three-dimensional (3D) radar can furnish all these data; i.e., azimuth, elevation, and range. This type of radar utilizes electronic scanning. One of the later classes of electronic scanning radars, the AN/MPS-23 (a component of the Missile Monitor FDS), provides three dimensional search. It supplies target azimuth, elevation angle, and range data simultaneously from a single antenna (transmitter and receiver) channel. The beam scans electronically in elevation while the antenna rotates in azimuth. The antenna frequency-scan operation is similar in principle to that of a slotted waveguide with the microwave energy radiated from the slots combining to form a beam. When the frequency is matched and phased with the distance between the slots, the direction of propagation is straight ahead. If the frequency of the energy is changed, relative phase differences are set up from one slot to the next, changing the direction of propagation accordingly. Continuous phase shifting is achieved by using variable frequency exciters in the transmitter. These exciters can be programmed digitally to provide various patterns of beams scanning in elevation. The AN/MPS-23 incorporates moving target indicator circuits that blank out returns from stationary objects. It is capable of azimuth*

sector scanning a, well as 6,400-mil rotational scan. It provides variable scan rates in elevation and azimuth and uses variable pulse repetition frequencies. The changing radiation frequency gives this radar inherent resistance against electronic jamming. (Ref: US Army Air Defense Digest, 1972)

MPS-24 Air Search
Frequency:
Power Output: 200 kw
Pulse Width: 5 µseconds
Pulse Repetition Rate: 60 pps
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer:
Notes: Used by Marine Corps

MPS-25 Tracker
Frequency: 5450 to 5825 MHz
Power Output: 1 Mw
Pulse Width: 0.25, 0.5, or 1 µseconds
Pulse Repetition Rate: 142-1364
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer: RCA
Notes: Mobile version of FPS-16.

MPS-30 Surveillance
Manufacturer: Hazeltine;
Notes: when trailer mounted, known as **TPS-25**

MPS-33

MPS-36 Tracking;
Power Output: 1Mw
Manufacturer: RCA or GE (?)

MPS-39 Multiple Object Tracking Radar (MOTR)
Frequency: 5400 to 5900 MHz
Manufacturer: GE(?)
Notes: Multiple Object Tracking Radar (MOTR) was a phased-array sensor capable of tracking forty objects in beacon or skin mode. Replaced air route surveillance radar system.

MPS-T9 – Early Warning/Ground Control Intercept Radar Threat Simulator
Manufacturer: DRS Technologies.

MSQ - Mobile; Special; Special

MSQ-1A - Matador Missile/Radar Bomb Scoring
Note: Primary use was in tracking and guidance for the TM-61C Matador Missile. It consisted of 3 Five ton vans. the radar van housed the MPS-19 radar. The MPS-19 had the capability of skin tracking from an APW-11 transponder. It aslo had a helical scan seach capability in that the antenna would start at zero elevation and automatically raise a few degrees every revolution then reset to zero and start again. The console van contained an analog computer and plotting board. It used range, azimuth and elevation informatiion from the radar to create a display on the plotting board. The third van contained spare parts and radio equipment.

MSQ-2 - Bomb Scoring Radar
Notes: **MPS-9** modified as **MSQ-2**, modified as **MSQ-39**, modified as **MSQ-46**

MSQ-18 - Tracking
Manufacturer: Hughes Aircraft Corp.

MSQ-35 -
Notes: Modified as **MSQ-77**
Manufacturer: Raythaeon Co.

MSQ-39 -
Notes: **MPS-9** modified as **MSQ-2**, modified as **MSQ-39**, modified as **MSQ-46**

MSQ-46 -
Notes: **MPS-9** modified as **MSQ-2**, modified as **MSQ-39**, modified as **MSQ-46**

MSQ-53 - phased array radar; search, target detection, track and identification;
Range: 100 Km
Notes: trailer mounted; tracks up to 100 targets; provides missile guidance data for up to nine missiles;

MSQ-77 - Ground Directed Bombing (GDB)
Range: 200 miles
Notes: Operation Rolling Thunder, Linebacker missions used precision GDB in Vietnam; after Vietnam, used for training; replaced by the AN/TPQ-43 "Seek Score" radar system;

MSQ-116 -
Notes: 5-ton truck transportable

MST - Mobile; Special or Combination; Transmitting

MST-T1A - MUTES
Manufacturer: General Dynamics
Notes: Sue Winkler first female assigned to the MUTES and first female to attend the first MUTES radar repair course.

Photos

MST-T1A(V) - MUTES
Notes: Smaller version of the MST-T1A MUTES

PPN - Portable; Radar; Navigation Aid

PPN-1 – Portable beacon,
Notes: For paratroopers; 1940-1945, 3,084 procured.

PPN-2 – Portable beacon,

Notes: for paratroopers; lighter version of PPN-1; 1940-1945, 4,392 procured.

PPS - Portable; Radar; Detecting, Range and Bearing, Search

PPS-3 Ground Movement
Manufacturer: Sperry

PPS-4 Ground Movement Doppler
Manufacturer: Sperry

PPS-5 Ground Movement
The AN/PPS-5 is a lightweight, transportable radar set. It is a partially transistorized set designed for detection of moving targets at short range.
Frequency: 16000 to 16500 MHz
Power Output: 1 kw
Pulse Width: 0.25 microseconds
Pulse Repetition Rate: 4,000 pps
Range: 6 km (personnel), 10 km (vehicles)
Vertical Coverage:
Horizontal Coverage:
Antenna:
  Speed:
  Beam Width: 20 or 45 mils horizontal; 62 mils vertical
Indicator Type and Quantity: A-scope for range, B-scope for azimuth
Weight: 196 pounds
Manufacturer: AIL
Notes: 125 pounds; Photo;

-5A-
Range: 10 km (personnel), 20 km (vehicles)
Notes: March 1975 Sole Source to Eaton Corp: 70 pounds;

-5B-
Notes: May 1978 Sole Source to Eaton Corp. - FMS Customer; September 1978 Sole Source to Eaton Corp; October 1989 Sole Source to Telephonics Corp. - FMS Customer;

PPS-6 Battlefield Surveillance
Description: lightweight pulse Doppler surveillance radar
Frequency: 9.0 to 9.5 GHz
Power Output: 100 watts peak
Transmitter: Magnetron, Varion
Pulse Width: 0.22 to 0.3µseconds
Pulse Repetition Rate: 1800 - 2200 pulses per second
Range: 50 - 3,000 meters; accuracy ±25 meters range, ±10 mils azimuth, ±15 mils elevation; resolution 50 meters in range, 360 meters in azimuth at 3,000 meters
Accuracy: ±25 meters
Vertical Coverage: +400 mils to -600 mils
Horizontal Coverage: 0 - 64 - mils, clockwise and counterclockwise
Antenna: Truncated parabolic, 12 inch diameter, gain 24.5 dB minimum
  Speed: 3 rpm
  Beam Width: 7° E-plane, 8° H-plane
Receiver: Minimum discernible Signal -95 dBm
Indicator Type and Quantity: Headset, Aural detection and identification, Range Indicator (mechanical counter) readout in meters.
Power: Self-contained 12V silver-zinc battery or external 12VDC source
Manufacturer: General Instruments Corp.
Notes: Weight: 35 pounds; tripod mounted.
Photos

PPS-9 Battlefield Surveillance
Manufacturer: General Dynamics
Notes: Weight: 10 pounds

PPS-11 Battlefield Surveillance
Manufacturer: General Instruments Corp.

PPS-12 Battlefield Surveillance
Manufacturer: General Instruments Corp.

PPS-15 Infantry Radar;
Power Output: 50mw; CW
Manufacturer: General Dynamics.

PPS-24 Perimeter De fence;
Manufacturer: ISC Defense

PPS-55 Battlefield Surveillance

SCR - Set, Complete Radio; or Signal Corp Radio

SCR-258 - Meteorological radar
Notes: subproject of the SCR-268; vertical dipoles; built 1939; azimuth and elevation antennas

SCR-258-T1 - Meteorological radar
Photo

SCR-268, T1, A, B, C - Search Light Control
Frequency: 205 MHz, 1½ meter wavelength
Power Output: 50 kw
Pulse Width: 7 to 15 µseconds
Pulse Repetition Rate: 4098
Range: 40,000 yards
Vertical Coverage:
Horizontal Coverage:
Antenna:
  Speed:
  Beam Width:
Indicator Type and Quantity:
Manufacturer:
Notes: Had a provision for data transmission to a searchlight or gun director; 1940-1945, 2,974 procured.
The SCR-268 Radar;
Photos

SCR-268-T1 -
Frequency: 10 MHz
Notes: Precursor to all Army radar; Conceived in February 1936; Delivered for Coast Artillery service tests November 1938; classified as a development type March 1939; Double-lobe tracking; last unit to contain heat tracking; rotated only in azimuth (not vertically); Transmitter and receiver on different mounts; only one made.

SCR-268-T2 -
Frequency: 205 MHz
Notes: The T1 was disassembled and used for parts on the T2 winter of 1938-1939; 15 May 1939 first unit turned over for testing, but never officially service tested;

SCR-268-T3 -
Notes: first unit to have transmitter and receiver on same mount; construction started March 1940; service tested 15-25 April 1940 at Fort Hancock; at the conclusion of the service tests, the T3 was designated Radio Set SCR-268, Required Type, Adopted Type, Standard Article, classified as Secret on 17 June 1940; at the same time the SCR-268-T1 was declared obsolete. February 1941 two sets to the Coast Artillery School at Fort Monroe, four to harbor defense and antiaircraft units, and eight to the Air Defense Command for the use of the 1st Aircraft Warning Company. In March, two set went to Fort Monmouth, 11 to harbor defense and antiaircraft units, and 12 to Aircraft Warning Companies. Reached a delivery rate of about one set per day in April 1941.

SCR-268-T4 -
Notes: first unit scheduled to have specially designed mounts for separate transmitter and receiver units; it was decided to concentrate on the single mounted T3, so one of the specially designed mounts was used for the T3.

SCR-268-T8 -
Pulse Repitition Rate: 205 pps
Range: (1) 0 to 40,000 yds. (2) 40,000 to 80,000 yds. (3) 80,000 to 120,000 yds.
Notes: Repackaging of the complete Radio Set SCR-268-C to a Radio Set SCR-268-T8; lightweight transmitter modulator; Eight tube transmitter; reduced the number of vehicles from five to two; 205 cycle keying adaptor made possible the selection of the the three ranges listed above.

SCR-268A -
Notes: a series of modification that were incorporated into the SCR-268 while the basic SCR-268 was in production. These field modification kits were to prevent the disruption of the manufacturing of the SCR-268. One prime mover pulling the antenna trailer; another prime mover pulling van trailer for power units.

Note

Actual changes to the B and C were minor. Clarified the supply of vehicles, maintenance equipment and spare parts.

SCR-268B -
Notes: One prime mover pulling antenna trailer; three prime movers for the rectifier, the antenna parts and test equipment, and the third as spare parts truck.

SCR-268C -
Notes: One trailer mount and one trailer van. Modified receiver with remote control tuning.

Built by	Type	Number Produced through 1944
Signal Corps Laboratories	SCR-268-T3	18
Western Electric Company	SCR-268	523 (including those later modified to SCR-268A
Western Electric Company	SCR-268B	1583
Western Electric Company	SCR-268C	850
Total	SCR-268-( )	2,974

SCR-270 -
Frequency: Optimum operating frequency: 105 to 107 MHz, 3 meter wavelength; Available operating frequencies: 102 - 112 MHz
Power Output: 100 kilowatts Peak

Pulse Width: 15 to 40 µseconds
Pulse Repetition Rate: 200 - 400 Hertz
Range: 0 - 200 miles
Vertical Coverage:
Horizontal Coverage:
Antenna: AN-140, 4-dipole high, 8-dipole wide, with screen reflector 1/4 wavelength behind the radiating elements
Speed:
Beam Width: 20 degrees between half-power points; Half power points lie at approximately 40 degrees elevation
Indicator Type and Quantity: Receiver-indicator Console BC-1239-A
Types of displays: PPI and A-scan
Ranges: 75 miles and 150 miles on the PPI, and 200 miles optimum on the A-scan
Range accuracy: Less than 3 miles
Azimuth accuracy: Less than 5 degrees
Manufacturer:
Notes: Models with elevation were the SCR-289; 530; and 531. 1940-1945, 397 procured. SCR-270 DA had 12-inch PPI and 12-inch scope added; Service trials late 1939; Officially adopted May 1940; Transportation: Antenna trailer, prime mover, operating truck and antenna storage truck; Photos

Service test held November and December 1939; Manufacturer Westinghouse Electric and Manufacturing Company; Official assignment of nomenclature 27 February 1939, and standardized with basis of issue of seven sets per Signal Air Warning Company 4 May 1940; First radar to use a duplexer;

Requirements:
"The apparatus must be so designed as to permit its movement from one position to another in permanent or semi-permanent defense installations, 24 hours being permissible for dismantling and re-erection exclusive of time for transport."

The various models of the SCR-270's are all designed for early warning use against aircraft. this is their primary purpose. Those sets which were equipped with the console conversion, could also be used as traffic-control stations. SCR-270s-B through SCR-270-DA had the transmitting, receiving and indicating equipment in trucks. Their antenna could be mounted on trailers (i.e. SCR-270-DA)

SCR-270-T1 –
Notes: Turned over to 1st Aircraft Warning Company early 1941; 2 built; assembled by the Signal Corps Laboratories; service test model;

SCR-270-A
Notes: 5 sets delivered to the 1st Aircraft Warning Company February 1941; assembled by the Signal Corps Laboratories; service test model;

SCR-270-B, -270 BA, -270-BB
Note: The -B is the basic set
Frequency:
Power Output: 100 kw peak
Pulse Width:
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width: Azimuth 28 degrees
Indicator Type and Quantity: -B, 5 inch A-scope; -BB has the receiver-indicator BC-1239-A added; -BA, does not have the BC-1239-A

Manufacturer: Westinghouse Electric and Manufacturing Company, 73 sets
Notes: Set that detected Japanese aircraft before Pearl Harbor attack at distance of 132 miles (received 1 August 1941)
The SCR-270-B consists of five road mobile units:
1. An operating van, containing the transmitter, water cooler, receiver, keying unit, and oscilloscope.
2. A power van, containing the gasoline engine, generator, and high-voltage rectifier.
3. A stake-body truck, used for packing and transporting the removable components of the antenna.
4. An antenna trailer, for transporting the tower portion of the antenna.
5. A prime mover, for hauling the antenna trailer.

Subsequent models incorporate seven types of modification:
1. Changes in the trailer or tower mounting the antenna
2. Changes in the r-f transmission line joining the transmitter, receiver and antenna
3. Changes in the antenna itself
4. Changes in the antenna position control system
5. Positional changes in the houses and in the trucks
6. Minor improvements not affecting the electrical functions of the transmitter, water cooler, rectifier, keying unit, oscilloscope, and receiver
7. A basic change in the indication system by the addition of the receiver-indicator console.

Stages of Improvement: Functionally, there were three transition phases through which the sets passed through in the successive modification.
1. The first phase incorporated the following:
a. Antenna AN-43, 9-dipole high, 4-dipole wide
b. Resonant r-f transmission line with one spark gap
c. Basic transmitting, keying and indicating equipment

2. The second transition phase incorporated the following:
a. Antenna AN-118-A, 8-dipole high, 4-dipole wide with screen reflector
b. The non resonant r-f transmission line with three spark gaps
c. The shielded rotating joint, antenna transmission-line coupling
d. The slightly modified transmitting, receiving, keying and indicating equipment
e. The improved Antenna Position Control containing provision for electrical data transmission

3. The third transition phase incorporated the following:
a. Antenna AN-140, 4-dipole high, 8-dipole wide
b. Receiver-indicator Console BC-1239-A
c. The improved antenna matching network
d. Improved data transmission kits
e. The same transmitting, keying and power supply as that incorporated in the second phase

SCR-270-C
Notes: SCR-270-B radar antenna on a 100 foot tower;

SCR-270-D
Frequency:
Power Output: 100 kw peak
Pulse Width: 10 to 30 µseconds
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna: Provision for use of a second antenna; non-resonant transmission lines for more stable operation; metal antenna replaced wooden antenna (reduced side lobes)

Speed:
Beam Width: Azimuth 28 degrees
Indicator Type and Quantity: 5 inch A-scope
Manufacturer:
Notes: Amplidyne added for antenna control;

SCR-270-DA
Frequency:
Power Output: 100 kw
Pulse Width:
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width: Azimuth 11 degrees
Indicator Type and Quantity: Indicator: 12 A-scope and 12 inch PPI
Manufacturer:
Notes: Designated the addition of the SCR-270-D modifications plus a SCR-527 receiver PPI indicator;

Number Produced	
Year	SCR-270 ( )
Before 1942	106
1942	148
1943	183
1944	66
Total	503

SCR-289 –
Notes: Updated SCR-270; Version of the SCR-270 designed to give elevation; See SCR-270

SCR-296-A - Harbor Defense, Fire Control
Note: The Army version of the Navy FC or Mark III. The set contained no suitable means for target acquisition; so that it was extremely difficult for the operators to pick up an assigned target if there are other targets nearby, without considerable coaching from surveillance radar. Resolution in azimuth was poor; and, as a result, interference between signals from two or more targets at approximately the same range may make it impossible to track anyone of them accurately. It is extremely difficult to spot shell splashes accurately, because of the poor resolution of the set. Another serious limitation of the SCR-296-A was its susceptibility to both electronic and "window" jamming, as a consequence of operating on the medium-long radar wavelength (43 centimeters).

SCR-516 - Ground Control Intercept;
Notes: An adaption of the SCR-268; intermediate range; converted at Signal Corps Laboratories

SCR-516A - Ground Control Intercept;
Notes: additional dipoles added and mount raised to accommodate the dipoles; only one (1) 516A built; converted at Signal Corps Laboratories

SCR-516B - Ground Control Intercept;
Frequency:
Power Output:
Pulse Width:
Pulse Repetition Rate:
Range:
Vertical Coverage:
Horizontal Coverage:
Antenna: Two antennas, one transmitting, one receiving with 4x18 dipoles; antenna mounted on 24 foot wooden tower
Speed: 6 rpm
Beam Width:
Indicator Type and Quantity: PPI
Manufacturer:
Notes: Further modified to use common antenna for transmitting and receiving; never put into production

SCR-516C - Ground Control Intercept;
Range: Field modified to 120,000 yard range; sixty-six modification kits issued

SCR-516D - Ground Control Intercept
Notes: Four sets converted at Signal Corps Laboratories from SCR-268B, otherwise the same as SCR-516C.

SCR-516E - Ground Control Intercept
Notes: Thirty-four converted SCR-268B's, done by Signal Corps Laboratories

SCR-527, A Ground controlled aircraft interception
Frequency: 209 MHz
Power Output: 225 kilowatts peak, 90 watts average
Pulse Width: 1 microsecond
Pulse Repetition Rate: 402 pps
Range: 0 to 240,000 yards
Vertical Coverage:
Horizontal Coverage:
Antenna: One dipole antenna mounted at focal point of parabolic reflector used for transmitting and receiveng. Controlled from the Controller position.
Speed:
Beam Width: 6 degrees
Indicator Type and Quantity: 7 inch Plan Position Indicator, Height Range Indicator
Manufacturer:
Notes: 1940-1945, 213 procured; Copied from British GCI. Radar consisted of three trailer, two power vans and two spare parts trucks.
Photos

SCR-530 –
Notes: Version of the SCR-270 designed to give elevation; See SCR-270

SCR-531 –
Notes: Version of the SCR-270 designed to give elevation; See SCR-270

SCR-539 –
Notes: A PPI indicator developed for use with the SCR-270 and SCR-271.

SCR-541 - Searchlight control radar;
Range: 15,000 yards
Notes: To be a lightweight version of the SCR-268; contract canceled November 1941;

SCR-545, A – Anti-aircraft fire control search and tracking;

Manufacturer: Western Electric Co., Bell Telephone Laboratories
Notes: 1940 -1945, 275 procured

SCR-547 – Anti-aircraft fire control range and elevation; stereoscopic height finder;
Frequency: 10 cm
Antenna: Two parabolic dish antennas
Notes: 1940-1945, 500 procured; Called Mickey Mouse because of its two ear like dish antennas; SCR 547 replaced by the SCR 584.

SCR-582 - Coast Defense Radar
Range: 25,000 yards
Indicator Type and Quantity: PPI
Notes: Was to be built for fixed installations, but provisions were made so that it could be truck mounted if the Army needed a mobile version.

SCR-582, Mark III -
Notes: 10 were built for the Army as mobile units, which later became the MPS-2.
Antenna: 10 diameter parabolic dish

SCR-582X - Coast Defense Radar
Notes: Also known as the Super X because of it's 10 foot dish antenna
Antenna: 10 diameter parabolic dish

SCR-584, -A, -B, –B1 Microwave fire control
Frequency: 2700 to 2900 MHz
Power Output: 210 kilowatts peak, 300 watts average
Pulse Width: 0.8 µseconds
Pulse Repetition Rate: 1707 pps
Range: 70,000 yards, Automatic Tracking 32,000 yards; accuracy ± 25 yards
Vertical Coverage: 300 yards to 10,000 yards
Horizontal Coverage: 360 °
Antenna: Dipole with helical scanning
Speed:
Beam Width: 4 degrees
Notes: Some SCR-584-A and - B1 models were not equipped with the necessary declutching mechanism for permitting continuous azimuth scanning. The dipoles antenna was not completely symmetrical; therefore, the antenna radiated a beam which was slightly offset from the axis of the reflector.
Indicator Type and Quantity: 12 inch PPI, 5 inch A-scope
Manufacturer: General Electric and Westinghouse Electric Manufacturing
Notes: 1940-1945, 1,610 procured; prototype was the **XT 1**; production started mid 1943. When the SCR-584 was first used as a mortar locator all plotting was performed manually. A sector was scanned until an echo appeared that was identified as being from a mortar shell, at which time the operator positioned the antenna directly over the spot where the echo first appeared and set the range gate to the proper range and waited. When the echo from a second shot appeared, the set was placed in automatic track status and the range, azimuth and elevation dials were read and recorded periodically on a given signal as the radar tracked the shell. The range and azimuth data were plotted to scale and a line drawn through the points produced a straight line which was considered as the ground projection of the trajectory. The range and elevation readings were used to compute the height and this information was plotted to an arbitrary scale perpendicular to the range-azimuth line. A parabola curve was then fitted to pass through the height plots and was extended back from the first point until it intersected the range-azimuth line. This was the position of the mortar. Any difference of altitude between the radar and the mortar position had to be accounted for either by adjusting the values of the height plots or by extrapolating below the range-azimuth line if the mortar position was below the radar plane and above the range-azimuth line if the mortar position was above the radar plane. This difference of altitude could be found from a contour map.

Photos

SCR-588 - Ground Control Intercept;
Notes: American copy, built in Canada, of the huge British CHL/GCI using long wavelengths;

SCR-598 - Gun laying
Notes: For coast defense, against surface vessels; similar to SCR-296 but of shorter range and greater accuracy for fire control purposes, directing gunfire against such small craft as motor torpedo boats; Later model became MPG-1;

SCR-602, Type 1 -
Frequency: 176 MHz, 212 MHz
Power Output: 100 K.W.
Pulse Width: 2 µseconds
Pulse Repetition Rate: 400 Hz
Range: 40 Miles (Three different sweep velocities create corresponding 10-mile, 40-mile and 100-mile ranges respectively.) Maximum Range: On flat site: 40 to 50 miles at 15,000 to 25,000) feet altitude. Minimum Range on flat site for low flying a/c 10 to 15 mi. Ceiling: About 25,000 feet altitude. Range Accuracy: Range to nearest mile. Azimuth accuracy 2 degrees.
Vertical Coverage:
Horizontal Coverage:
Antenna: Four horizontally polarized Yagi antennas.
Speed: Continuous at about 6 RPM.
Beam Width:
Indicator Type and Quantity: 9 inch PPI, Height A-scope
Manufacturer: Research Enterprises Limited (Canada)
Notes: 1,200 pounds; First 25 of manufactured were exact copies of the British LW prototype; last copy of a British radar and the last copy of a long range set (wavelength about one and a half meters) which the Signal Corps produced for the AAF. Equipment rack mounted in two adjoining frame racks. The receiver rack houses power equipment in the bottom, the main PPI and ASV receiver in the central portion, with the PPI power unit and indicator unit at the top. The adjoining transmitter rack continues up in a tapered tower through the tent top to support the rotatable Yagi antenna. The lower part of this rack houses the transmitter, T. and R switching, unit, inductive coupling unit to antenna, and the antenna rotating gear box. Power supply: 6 h.p. gasoline engine drives two 700 watt, 80-volt, 1,000-cycle alternators, and one (1) 500-watt 24-volt d.c. generator. The d.c. generator supplies the alternator fields, blowers, lights and charges batteries. One (1) alternator supplies the transmitter only.

Documents: Results of SCR-602 Flight Tests 22 April 1943

Photos

SCR 602 Type 2 -

SCR 602 Type 3 -

SCR 602 Type 4 -

SCR 602 Type 5 -

SCR 602 Type 6 -
Notes: Next 250 built after the Type 1

SCR 602 Type 7 -
Manufacturer: General Electric
Notes: 250 pound parachute-able set

SCR 602 Type 8 -
Frequency:
Power Output:
Pulse Width:
Pulse Repetition Rate:
Range: 40 Miles
Vertical Coverage:
Horizontal Coverage:
Antenna:
Speed:
Beam Width:
Indicator Type and Quantity:
Manufacturer:
Notes: Signal Corps developed, proved to be the best; receiver, display and high power transmitter in a single unit measuring 42 inches by 20 inches by 20 inches; became AN/TPS 3. Included new IFF Interrogation equipment, built in a separate case.

SCR 602-A

SCR-615 - Ground Control Approach;
Frequency: 2,750 to 2,800 MHz, 10.7-cm microwave radar
Power Output: 600 kilowatts peak, 480 watts average
Pulse Width: 1 microsecond
Pulse Repetition Rate: 800 pps
Range:100 miles maximum, 2 - 5 miles minimum
Vertical Coverage: 107 to -3 degrees
Horizontal Coverage: 360 degrees
Antenna: A waveguide radiator with a target and an 8-foot parabolid reflector mounted so that the whole assembly can be rotated in azimuth and tilted in elevation. Conical scanning accomplished by rotation of waveguide radiator.
Speed:
Beam Width: 6 degrees
Indicator Type and Quantity: 7 inch PPI, 5 inch A-scope, 5 inch R-scope

SCR-615-B - Air Transportable Height Finder
Notes: The Radio Set SCR-615-B is a fixed air transportable height finding ground radar designed for height finding and low angle coverage in conjunction with ground controlled interception. It is essentially a fixed version of the SP-1M. Although physically similar to the SCR-615-A, the circuits are completely redesigned.
Frequency: 2000 MHz
Power Output: 750 kilowatts peak
Pulse repetition rate: 400 pps
Pulse width: 1.5 microseconds
Antenna: Continuous rotation at any chosen angle of elevation; Sector sweep at any chosen angle of azimuth or elevation where the limits of the azimuth sector can be chosen; Sector sweep in both azimuth and elevation where the azimuth sector can be from 1 degree to 360 degrees, but the elevation sector is preset at 12 degrees; Sector sweep in both azimuth and elevation where the sectors can be continuously adjusted.

SCR-627, A -

SCR-636; A -

SCR-648 -

SCR-668 - Search Light Control sets;
Notes: Enclosed in tents

SCR-668-T1 -
Notes: Antenna arrays mounted over the face of searchlight

SCR-668-T2 -
Frequency: 30-cm band;
Antenna: Two parabolic cylinder reflectors mounted separately from searchlight

SCR-668-T3 -
Frequency: S-band
Notes: transported on separate searchlight trailer; standardized in January 1944 as the TPL-1

SCR-668-T4 -

SCR-668-T5 -

SCR-668-T6-

SCR-682-A – Harbor Surveillance
Frequency: 2,800 MHz
Power Output:
Pulse Width:
Pulse Repetition Rate:
Range: 500 yards to 240,000 yards
Accuracy:
10,000 yard range scale: ±100 yards
40,000 yard range scale: ±500 yards
160,000 yard range scale: ±2,500 yards
240,000 yard range scale: ±5,000 yards
Vertical Coverage: Manually adjustable from -5 degrees to +25 degrees
Horizontal Coverage: 360 degrees, ±1 degree accuracy
Determination of direction: Read on PPI scope azimuth dial in number of degrees from true South.
Antenna: Continuous rotation of spinner assembly at 10 rpm (6 rpm for serial numbers 1 - 6).
Speed:
Beam Width:
Indicator Type and Quantity: 7 inch PPI
Transportation: Three 2 1/2 ton, 6 X 6, LWB cargo trucks or two 2 1/2 ton, 6 X 6, LWB cargo trucks, and two 1 ton trailers. Erection time was 3 to 5 hours by a trained crew of five.
Manufacturer:
Notes: 1940-1945, 140 procured; Improved SCR-582. Unit could be adjusted to search for aircraft, but ship detection capability was lost. Used with 155 mm Gun Battalions.

SCS-768 - Search Light Control
Notes: Experimental, patterned after the British "Wigwam" radar;

SCR-784 - Fire Control
Frequency: 2,800 MHz
Power Output:
Pulse Width: 0.8 µseconds
Pulse Repetition Rate: 1707 pps
Range: 9,000 yards
Vertical Coverage: 300 yards to 10,000 yards
Horizontal Coverage:
Antenna:
Speed:
Beam Width:

Indicator Type and Quantity: Display 7 inch PPI and two 3 inch CRT's for range determination
Manufacturer:
Notes: Trailer mounted; 1940-1945, 80 procured; lightweight version of the SCR-584. It is the same as the -584 electrically, but weighs only half as much and has a lower silhouette. Also has automatic tracking in range as well as in elevation and azimuth.
Photos

SCR-802 -
Frequency: 10-cm
Range: 15 miles
Antenna: Parabolic reflector
Notes: Lightweight portable Radar primarily for use at night; Setup time 15 minutes; canceled because of inability to secure a suitable manufacturer for production, 12 August 1943

SCR-825 -

SO

SO-12-N - Seacoast Surveillance Radar (Navy)
Note: Used by mobile 155 mm Gun Battalions
Indicator Type and Quantity:Display 7 inch PPI; VG; the VG remote indicator was a projection-type, dark-trace oscilloscope. It included a special cathode-ray tube, called a Skiatron, on which signals appear in a dark magenta color against a light background. The image was projected, through a magnifying lens system, upon the under surface of a ground-glass screen about 30 inches in diameter. A translucent overlay bearing a map of the local area could be placed over the ground-glass screen, for maintaining a plot of target courses. The VC unit, possessed a real advantage as a, remote indicator for surveillance radar, in Group or Harbor Defense Command Posts and in Harbor Entrance Control Posts. It could be viewed in a fully lighted room, in contrast to the usual types of radar indicators, which had to be operated in subdued light or total darkness. Thus anyone could observe the screen at any time without waiting until their eyes fully adapted to a radical change in illumination levels. Since the screen of the Skiatron tube is of the long-persistence type, images of successive positions of the target signal remained for a long time, leaving a continuous "track.

SP

SP-1M - Mobile Tracking Radar
Radio Set SP-1M is a Marine Corps mobile ground radar designed for fighter direction in conjunction with ground controlled interception. It is essentially a mobile version of the SCR-615-B. Entire SP-1M equipment necessary for fighter control is mounted in two trailers which contain the radar, communications, identification, plotting table, and power supply.
Frequency: 3000 MHz
Power Output: 600 kilowatts
Pulse repetition rate: 600 or 120 pps
Pulse Width: 1 or 5 microseconds
Antenna: 8 foot parabolic antenna; conical scan
Beam width: 3 degrees azimuth, 4 degrees in elevation at half power points.
Bearing Accuracy: ±1/4 degrees
Elevation accuracy: ±700 feet
Indicator Type and Quantity: A-scan with 100 mile range sweep; R-scan with a 5 mile

sweep; PPI with 4, 20, 50 and 100 mile ranges; Projection PPI for use as a plotting table
Notes: Identification equipment: Navy type BN and BO IFF

Photos

TLQ - Ground, Transportable; Countermeasures; Special

TLQ-17 -

TPB - Ground, Transportable; Radar; Bomb Scoring

TPB-1 B - Ground Directed Bombing System
The AN/TPB-1B is a highly mobile ground directed bombing system designed to provide night/all weather guidance to tactical strike, reconnaissance, and airlift aircraft. The system consists of a computer controlled radar set operating in conjunction with an aircraft equipped with a beacon (Motorola SST-181XE or equivalent). The system operating controls can be housed in a mobile shelter for relatively stable operations or in a collapsible tent for the mobile configuration. The radar is a small, relatively lightweight unit that can quickly be assembled or disassembled in the field. All components of the system can be packed inside the shelter for ease in transportation. The AN/TPB- 1B is transportable by XM-720 transport mobilizers drawn by M-35 (2 1/2 ton) truck and can also be transported by cargo, helicopter, and man-pack methods. The system itself utilizes solid state circuitry to provide compactness and increased system reliability. The AN/TPB-1B utilizes a pencil beam radar operating in the I-band. The radar functions to interrogate the beacon, automatically track the aircraft, provide range/azimuth /evaluation data to the self-contained digital computer that computes the ballistics problem, and display aircraft position data to the operator. All system modes of operation are selected through the four units of the radar control assembly housed in the shelter and connected to the antenna by a 100 foot cable.
Capability: Manual or Automatic scan 360 degrees; Only one beacon equipped aircraft can
be controlled at a time.
Frequency Range: Transmitter and Receiver frequency 8.8 to 9.6 GHZ (Tunable).
Power Output: 100 kW peak
Weight: 2514 pounds
Manufacturer: Sierra Research Corporation

TPG - Ground, Transportable; Radar; Fire Control or Searchlight Directing

TPG-1 -Costal Gun Layer
Frequency: 10,000 MHz, 3-cm;;
Pulse Width: 0.25 µmicroseconds

TPL - Ground, Transportable; Radar; Searchlight Control

TPL-1 - Searchlight Control
Description: Radar Set AN/TPL-1 is a mobile, compact, lightweight radar unit designed for use with an antiaircraft searchlight. In all conditions of normal operation the unit will be carefully sited and accurately oriented. If desired, however, the unit may be used to give indications of approaching aircraft when the equipment is being transported in convoy. The general functions of the equipment are as follows:
a. To search the sky for enemy aircraft and, when found, to indicate their position. Position data is furnished as slant range (distance in yards) and as azimuth and elevation

(angular directions in mils). The radar set also computes target altitude in feet. The altitude may be measured with either the radar set or sea level as the reference. The maximum slant range of the equipment is 60,000 yards.
b. To furnish target-position data to an antiaircraft searchlight so that the beam of the searchlight, when it is turned on, will strike the aircraft.
c. To search for a second target while the searchlight follows the original target.
Frequency:
Power Output:
Pulse Width:
Pulse Repetition Rate:
Range: 35 miles, 60,000 yards
Vertical Coverage: -100 mils to +1,600 mils in manual mode, 125 mils to 825 mils in steps of 100 mils in automatic scan
Horizontal Coverage: 360 degrees
Antenna: 48 inch Parabolid, with spinning dipole antenna
Speed: 7 rpm in scanning mode
Beam Width:
Indicator Type and Quantity: 7-inch PPI, 5-inch A-scan, two 2-inch A-scan tracking indicators for elevation and azimuth, and four azimuth and elevation data dials
Weight: 2,550 pounds
Manufacturer:
Notes: Lightweight microwave; 60 inch Searchlight Control; developed from SCR-668. 1940-1945, 383 procured. The main console cabinet holds all the other components of the radar including the four (4) wheels on which the cabinet is mounted. The radar is meant to be transported in another trailer as the wheels on the radar are not spring supported. Radar searches and tracks targets.
Photos

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